INCH-POUND

MIL-DTL-46119D
27 July 2026
SUPERSEDING
MIL-S-46119C(MR)
26 October 1988

# DETAIL SPECIFICATION

# STEEL FORGINGS, TUBULAR PARTS FOR CANNON

This specification is approved for use by all Departments and Agencies of the Department of War

## 1. SCOPE

1.1 Scope. This specification covers steel forgings that are produced by a hot deformation process such as hammering, pressing, upsetting, extruding, or any combination of these processes (see 6.1). This specification includes the minimum essential Engineering and Packaging Requirements and the necessary Quality Assurance Provisions to determine that these requirements have been met.

## 2. APPLICABLE PUBLICATIONS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: The U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny, NJ 07806-5000 or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A AREA FORG

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2.2 Government documents.

2.2.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF WAR SPECIFICATIONS

MIL-DTL-13931 - Cannon, General Specification For

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

DEPARTMENT OF WAR STANDARDS

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

MIL-STD-2073-1 - Standard Practice for Military Packaging

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

SMCWV FORM 1155 - Record of Treatment and Test of

(Copies of this document should be obtained from the contracting activity, or as directed by the contracting activity.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM A751 - Standard Test Methods and Practices for Chemical Analysis of

ASTM D3951 - Practice for Commercial Packaging

ASTM E8/E8M - Standard Test Methods for Tension Testing of Metallic



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Materials

| | | |
|---|---|---|
| ASTM E23 | - | Notched Bar Impact Testing of Metallic Materials |
| ASTM E340 | - | Method for Macroetching Metals and Alloys |
| ASTM E381 | - | Standard Method of Macroetch Testing Steel Bars, Billets, Blooms, and Forgings |

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 General requirements. Steel forgings shall be in accordance with the applicable paragraphs of MIL-DTL-13931 as incorporated in this document, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.4).

3.2 First article. The contractor shall submit a first article sample unless it is specifically waived in the contract (see 4.2, 6.2, and 6.3). No first article requirements shall be waived without review and approval by the procuring contracting officer.

3.2.1 First article waiver. The requirement for a first article may be waived on a new procurement of forgings of similar design and strength level to those previously procured from the same contractor (see 6.13).

3.2.1 First article residual metal. Metal removed from the forging prolongations (see 6.5.3) also known as residual metal from the first article forgings shall be retained until the acceptance of the first article (see 6.2).

3.3 Chemical composition. Unless a particular composition of steel is specified on the drawing, the contractor shall select a steel composition which when heat treated shall meet the mechanical properties specified on the applicable drawing, or in the contract. (see 4.3.1 and 6.7).

3.4 Processing controls.

3.4.1 Orientation of forgings to ingot. Forgings shall be produced so that the longitudinal axis of the forging is parallel or coincident with the longitudinal axis of the ingot. (see 4.3.2).

3.4.2 Straightening. Bows not exceeding 0.005 inch (arc to chord) per inch of length over which the bow extends may be removed by cold straightening after the final tempering heat treatment. If any bow exceeds 0.005 inch per inch, it shall be removed by hot straightening. The minimum temperature of the forging during the hot straightening operation is 600 degrees F (316 degrees C). The maximum temperature the forging can be heated to prior to the straightening operation is the final tempering temperature minus 50 degrees F (28 degrees C). Each forging that is hot straightened shall be retested for impact resistance properties after hot straightening.



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3.4.3 Welding. Forgings shall not be welded except on prolongations. All welds shall be in such a position so as to not affect the finished tube or test metal.

3.5 Heat treatment. Heat treatment shall be performed on the whole forging (see 6.5.2), never on a portion only, and shall consist as a minimum of austenitizing, agitated liquid quenching and tempering. Forgings for tubes with 57 mm (2.24 inches) finished bore diameter and larger shall be rough bored and rough turned to the applicable forging drawing dimensions before heating and quenching. Unless otherwise allowed, the liquid coolant shall pass freely through the bore during the quenching process. Forgings for tubes with finished bore diameter under 57 mm (2.24 inches) may be quenched solid at the option of the contractor. Tempering shall immediately follow quenching and shall consist of heating the entire forging to a temperature not less than 1000 degrees F (538 degrees C) for yield strength ranges up to and including 205,000 pounds per square inch (psi), and holding at the temperature for a time appropriate to the thickness. A zone-controlled tempering treatment employing different temperatures at opposite ends of the forging (differential tempering) may be used at the option of the contractor. Whenever differential tempering is employed, or whenever forgings are retempered only, temperatures of 1000 degrees F (538 degrees C) minimum are required, as noted, unless otherwise allowed by the procuring contracting officer, (see 6.2). When stress relieving is required after heat treat and temper, the stress-relief temperature shall be a minimum of 50 degrees F (24 degrees C) below the tempering temperature (see 6.8).

3.6 Mechanical properties.

3.6.1 Yield strength. Yield strength shall be as specified on the applicable drawings, or in the contract or purchase order. In addition, the spread in yield strength within any disc or between any two adjacent discs shall not exceed 4%, or 6,000 psi, whichever is greater (see 4.3.5.1).

3.6.2 Reduction of area. Unless specified on the applicable drawings, or in the contract or purchase order, the minimum percent reduction of area for individual specimens shall be as specified in Table I (see 4.3.5.2), for the average yield strength obtained (see 6.5.1).

3.6.3 Charpy V-notch impact resistance. Unless specified on the applicable drawings, or in the contract or purchase order, the minimum value for individual Charpy V-notch impact resistance specimens shall be as specified in Table I (see 4.3.3.3.2), for the average yield strength obtained (see 6.5.1).

TABLE I. Mechanical properties.

| **Yield strength (psi, 0.10 percent offset)** | **Minimum reduction of area (single value) Percent (%)** | | **Minimum Charpy V-notch impact resistance at -40F ± 2F (-40 ± 1C) Single value foot-pounds)** | |
|---|---|---|---|---|
| | **Transverse** | **Longitudinal** | **Transverse** | **Longitudinal** |
| 120,001 - 130,000 | 45 | 50 | 40 | 50 |
| 130,001 - 140,000 | 40 | 45 | 35 | 45 |
| 140,001 - 150,000 | 35 | 40 | 30 | 40 |
| 150,001 - 160,000 | 35 | 40 | 25 | 35 |
| 160,001 - 170,000 | 30 | 35 | 20 | 30 |
| 170,001 - 180,000 | 30 | 35 | 20 | 25 |
| 180,001 - 205,000 | 30 | 35 | 20 | 25 |
| Greater than 205,000 | (see applicable forging requirements, see 6.12) | | | |



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3.7 Macrostructure. Macrostructure of forgings shall be equivalent to or better than Standards S-2, R-2, and C-3 of ASTM E381, Plate 1 and shall also meet the requirements of Plate II. Standard C-3 only applies when the forging is heat treated solid (see 3.5).

3.8 Dimensions and surface roughness. Dimensions and surface roughness, and dimensional tolerances shall be as specified on the applicable forging drawing.

3.9 Identification marking. Each forging shall be marked on the breech end, by metal-die stamping, with the heat, forging serial number, drawing number, and the manufacturer's identification (see 4.7). When special identification marking is required, (see 6.2).

3.10 Workmanship. The forging shall be free from defects such as laps, seams, cracks, etc. which cannot be removed by machining the forging to its finished dimensions. The existence of these defects shall cause the forging to be rejected even if all other requirements of this specification are met (see 4.8).

## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Quality conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 General. Unless otherwise specified in the contract, (see 6.2), a first article sample shall be submitted for inspection. The first article shall be representative of the production processes to be used during quantity production. (see 6.3)

4.2.2 First article sample. Unless otherwise specified in the contract, the first article sample shall consist of one forging (see 6.2 and 6.3).

4.2.2.1 Inspection provisions. The first article shall be subjected to quality conformance inspection (see 4.5), additional inspections specified herein, and the provisions of the contract. A minimum of one (1) forging in the sample shall be destructively tested by taking specimens at each end and at ¼, ½, and ¾ of the length of the forging unless otherwise specified. The number of specimens and tests required for each of these locations are listed in Table II.

4.2.2.2 Number of specimens and tests. Unless otherwise specified on the applicable drawing or in the contract or purchase order (see 6.2), the number of specimens and tests for the individual or multilength forgings shall be as shown in Table II. (see 6.3)



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TABLE II. Number of specimens for each test.

| Test | Number of Specimens | |
|---|---|---|
| | Individual Forgings | Multilength Forgings |
| Macrostructure | 1 - each end | 1 - at each end and<br>1 - at each junction between adjoining lengths |
| Tension | 2 - each end | 2 - at each end and<br>2 - at each junction between adjoining lengths |
| Charpy V-notch | 2 - each end | 2 - at each end and<br>2 - at each junction between adjoining lengths |

4.3 Quality conformance inspection. All forgings shall be subjected to quality conformance inspections as specified in Table III which lists inspection methods and their corresponding requirements.

TABLE III. Quality conformance inspection.

| Examination or Test | Requirement Paragraph | Examination or Test Paragraph |
|---|---|---|
| Chemistry | 3.3 | 4.3.1.1 |
| Orientation of forgings to ingot | 3.4.1 | 4.3.2 |
| Yield Strength | 3.6.1 | 4.3.3.3.1<br>4.3.3.4.1<br>4.3.5.1 |
| Reduction of Area | 3.6.2<br>(Table I) | 4.3.5.2 |
| Charpy impact resistance | 3.6.3<br>(Table I) | 4.3.3.3.2<br>4.3.3.4.2<br>4.3.5.3 |
| Macrostructure | 3.7 | 4.3.4.1<br>4.3.4.2 |
| Identification marking | 3.9 | 4.7 |
| Workmanship | 3.10 | 4.8 |
| Packaging | 5.0 | 4.9 |

4.3.1 Chemistry. One sample for chemical analysis shall be taken from each forging from any of the discs prepared for mechanical tests.

4.3.1.1 Chemical Testing. Chemical testing shall be in accordance with the applicable method specified in ASTM A751. The testing shall be conducted for the chemicals identified on the SMCWV FORM 1155 (see 6.7).



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4.3.2 Forging orientation to ingot. A verification shall be performed for each forging confirming the longitudinal axis as parallel with the longitudinal axis of the ingot in accordance with ASTM E340 and ASTM E381 macroetch methods.

4.3.3 Mechanical properties.

4.3.3.1 Location of discs. All discs from quenched and tempered forgings shall be located at a distance of at least one wall thickness from the end of the forging (see 6.5.2). After discs have been taken, in accordance with the above requirements, additional thermal treatment of the forging shall be limited to stress relieving, retempering, hot straightening, or retempering and hot straightening (see 3.4.2). Additional discs from each end are required adjacent to the previous discs to each provide two tension and two Charpy impact specimens after any forging has been re-heat treated or hot straightened. Test values shall comply with the applicable forging drawing and Table I.

4.3.3.2 Location of test specimens.

4.3.3.2.1 Transverse. Tension and Charpy V-notch impact test specimens shall be taken from the location indicated on the applicable drawing. Specimens shall be taken so that the longitudinal center lines of the specimens are in a plane approximately perpendicular to the longitudinal axis of the forging. Charpy V-notch specimens shall be notched on the side nearest the bore surface. Breech end and o'clock positions shall be maintained for each disc. Tension specimens shall be taken at the 12 and 6 o'clock positions. Charpy impact specimens shall be taken at the 3 and 9 o'clock positions. If additional specimens are required, they shall be taken from the same disc or an adjacent disc when sufficient material is not available.

4.3.3.2.2 Longitudinal specimens. When the forging design precludes the use of transverse specimens, longitudinal specimens shall be used, in which case the specimens shall be taken with their longitudinal centerlines approximately parallel to the longitudinal axis of the forging. Charpy specimens shall be notched on the side nearest the bore surface. The positions of the tension and Charpy specimens shall be the same as in 4.5.1.2.1.

4.3.3.3 Test specimens.

4.3.3.3.1 Tension. Tension test specimens shall be machined to the form and dimensions of the standard 0.250, 0.350, and 0.500 inch diameter test specimen with 1.000, 1.400, and 2.000 inch gauge lengths respectively, as shown in ASTM E8.

4.3.3.3.2 Charpy V-notch impact. Charpy V-notch impact specimens shall be machined to the form and dimensions of the standard 0.394 inch by 0.394 inch by 2.165 inch specimen shown in ASTM E23, unless otherwise indicated in the engineering drawing.

4.3.3.4 Test Methods.

4.3.3.4.1 Tension. Yield strength shall be determined by the offset method in accordance with ASTM E8. The limiting offset shall be 0.10 percent (0.001 inch per inch of gauge length). The rate of stress application when determining the yield strength shall not exceed 100,000 pounds per square inch per minute.



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4.3.3.4.2 Charpy V-notch impact. Charpy V-notch impact tests shall be conducted in accordance with ASTM E23. The temperature of the test specimen at impact shall be minus 40 ± 2 °F or minus 40 ± 1 °C.

4.3.4 Macrostructure.

4.3.4.1 Test discs. Discs shall be cut from each end of the forging for macroscopic inspection. The discs shall be taken from metal at least one wall thickness from the end of the forging as originally heat treated. At the option of the contractor, the discs may be those taken from each end for mechanical testing. Both discs shall meet the requirements of 3.7.

4.3.4.2 Test method. Discs shall be prepared and examined in accordance with ASTM E340 and ASTM E381 to determine compliance with 3.7.

4.3.5 Interpretation of test results.

4.3.5.1 Yield strength. The forging shall have met requirements for yield strength if all values of yield strength fall within the limits prescribed by the applicable drawings and the requirements of 3.6.1.

4.3.5.2 Reduction of area. The forging shall have met the requirements for reduction of area if all values of the reduction of area are equal to or greater than that shown in Table I. The minimum percent reduction of area is shown opposite the applicable yield strength range (see 3.6.2).

4.3.5.3 Charpy V-notch impact resistance. The forging shall have met the requirements of Charpy V-notch impact resistance if all values of the impact resistance are equal to or greater than that shown in Table I. The minimum foot-pounds resistance is shown opposite the applicable yield strength range, into which the average yield strength falls (see 3.6.3 and 6.5.1).

4.4 Rejection. Forgings which fail to meet the requirements of this specification when tested in accordance with 4.5 or any other required inspection specified in the contract or purchase order shall be rejected.

4.4.1 Local defects. If rejection of a forging can be traced to the presence of allowable defects (see 3.7) in the failed test specimen, the test values obtained on that specimen may be disregarded subject to the approval of the contracting officer. Two wake specimens (see 6.5.6) shall be substituted for the defective specimen. Only one such substitution is permitted per forging. When the wake specimens have been substituted, failure of the forging to meet specification requirements based on test results from the wake specimens shall be cause for rejection of the forging, whether or not the wake bars also exhibit defects.

4.4.2 Defective machined specimens, faulty test equipment, and faulty testing methods. If either a defectively machined test specimen, faulty test equipment, faulty operation of the testing equipment, or recording device results in the rejection of a forging, the test values obtained from such specimens may be disregarded subject to the approval of the contracting officer, and another specimen from the same forging location may be substituted for each such specimen.



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4.4.3 Treatment for retest. Forgings that do not meet the yield strength, reduction of area or Charpy V-notch resistance requirements of this specification except as noted in 4.5, shall be reheat treated before retesting. Reheat treatment may consist of retempering or reaustenitizing, agitated liquid quenching, and retempering. All retempering without reaustenitizing shall be at a temperature equal to or greater than the original tempering temperature for the forging.

4.5 Check tests. From any lot of forgings found acceptable under prescribed inspection, the Government may designate any forging as a check forging. Such a forging shall be tested at a facility to be designated by the Government, and shall be subjected to any or all tests specified herein. If the forging fails to comply with the requirements if the applicable engineering drawing and Table I, two additional forgings shall be tested. If both forgings comply with all requirements, the lot shall be accepted. If either forging fails to comply with all the requirements, the lot shall be rejected.

4.5.1 Check test residual metal. Residual metal from any check forgings shall be retained until the check forgings are either accepted or rejected (see 6.2).

4.6 Final acceptance. Final acceptance of forgings shall be made when the final inspection satisfactorily establishes their compliance with the requirements of this specification.

4.7 Identification marking. Each forging shall be visually examined for identification marking. Failure to comply with 3.9 shall be cause for rejection.

4.8 Workmanship. Each forging shall be visually examined for workmanship. Failure to comply with 3.10 shall be cause for rejection.

4.9 Packaging. Packaging, packing and marking for shipment shall be visually examined. Failure to comply with the requirements of Section 5 shall be cause for rejection (see 5.1).

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoW or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity (see 4.9).

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The steel forgings specified herein are intended to be used as tubular parts of military cannon such as tubes, jackets, and liners.



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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specification.

b. Issue of military specification to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Instructions for the submission of a first article, including the number of forgings, unless the first article is waived (see 3.2, 4.2.2, and 6.3).

d. Residual metal (see 3.2.1 and 4.7.1).

e. Steel composition, if specified (see 3.3).

f. Heat treatment to be used if required (see 3.5).

g Applicable drawings or sketches showing locations and numbers of test specimens, and mechanical property requirements (see 3.6 and 4.4).

h. Special identification marking not covered in 3.9.

i. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

j. All inspection and test data that is required including any additional or extended examinations and tests beyond the scope of this specification.

k. At whose expense a retest may be performed.

l. Inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified.

m. Availability of inspection equipment from the Government.

n. Extent of contractor's responsibility for Government furnished and for contractor-required final inspection equipment.

o. Applicable acceptance test procedures.

p. Packing and marking for shipment (see 5.1).

q. If alternate sampling is indicated (see 4.1 and 6.11).

r. Waivers and deviations (see 6.4 and 6.13)

s. Heat treatment procedure (see 6.8)

t. Mechanical properties Record of Treatment and Test (SMCWV FORM 1155) (see 6.12)

u. Responsibility for inspection (see 6.10)

v. Responsibility for compliance (see 6.11)

6.3 First article. When a first article inspection is required, the procuring contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first (how many) production items, a standard production item from the contractor's inventory, and the number of items to be tested as specified in 4.3. The procuring contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the



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Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, should furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation. (see 6.2)

6.4 Waivers and deviations. The procuring contracting officer should coordinate all requests for waiver or deviation to this specification with Benet Laboratories and the Quality Directorate at Watervliet Arsenal (see 6.2).

6.5 Definitions.

6.5.1 Average yield strength. The average yield strength of the forging under consideration is the arithmetical mean of all yield strength values obtained from the same end at the stage of heat treatment at which tests are being performed.

6.5.2 Forging. A forging for purposes of inspection and test is defined as the forging, the length of which is shown on the applicable rough machined forging drawing, plus any prolongations as defined in 6.5.3 which are integral parts of the piece as heat treated. In the case of forgings formed and heat treated in multiple lengths, one set of specimens (in addition to the set of specimens obtained from each end of the multiple forgings) should be obtained from the test metal between each pair of adjoining forgings to be applicable to both forgings. Provisions should be made for identification of test specimens with the particular forging within a multiple forging.

6.5.3 Prolongation. A prolongation, for purposes of inspection and testing is defined as an extension beyond the rough machined forging length. The diameters and mass distribution of the prolongation should not be less than those of the forging at a point coincident with the end of the rough machined forging, from which it extends.

6.5.4 Retest. A retest is defined as any test made after the results of the original test are known.

6.5.5 Test metal. Same as Prolongation (see 6.6.3).

6.5.6 Wake bar. A wake bar is a test specimen which is substituted for another test specimen which has been discarded because of the presence of local defects caused by or overlooked in the preparation of the specimen. The wake bar should be taken as close as possible to the site of the original test specimen.

6.6 Finish-machined contour. The contour of the finish-machined tube should be indicated on the forging drawing or on a separate drawing furnished by the procuring contracting officer.

6.7 Chemical composition. The contractor should submit their proposed chemistry with their bid for forgings for informational purposes only. The contractor should furnish a certified heat analysis for each forging. ASTM A751 should be consulted for a list of available methods.



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6.8 Heat Treatment. For quenched and tempered forgings with yield strength ranges up to and including 205,000 psi, the contractor should submit their proposed heat treatment procedure with their bid for forgings for informational purposes only (see 6.2). For alloys with a yield strength greater than 205,000 psi, the heat treatment should be as agreed upon between the contractor, the procuring contracting officer, and the design authority.

6.9 Inspection and quality assurance. Certification of a contractor's quality system or program is not required by the Department of War if ISO or ANSI/ASQC standards are used.

6.10 Responsibility for inspection. Unless otherwise specified in the contract, the contractor should be responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements (see 6.2).

6.11 Responsibility for compliance. The inspection set forth in this specification should become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification should not relieve the contractor of their responsibility for assuring that all supplies submitted to the Government for acceptance comply with all requirements of the contract or purchase order. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material. If alternate sampling is indicated, or requested, the procuring activity should refer to MIL-STD-1916 and ANSI/ASQ Z1.4 (see 6.2).

6.12 Mechanical properties records. The results of all tests taken on each forging should be recorded. These records should include results of tests taken before and after retreatment and the discarded results of any tests with the reasons given for the discard. All heat treatment cycles and results of all tests taken on each forging should be reported on SMCWV FORM 1155, entitled; Record of Treatment and Test, or its current equivalent. The contractor should record the information needed to comply with this specification to enable the Government representative to verify and sign SMCWV FORM 1155 (see 6.2).

6.13 First article waiver. The waiver and determination of similarity should be made by the procuring contracting officer (see 3.2.1, 6.2, and 6.3).

6.14 Metric units. When metric dimensions are required, units for inch, foot, foot-pounds, and feet per second may be converted to the metric equivalent by multiplying them by the following conversion factors listed in Table V.



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TABLE V. Conversion factors.

| English | Multiply by | Equals | Metric SI unit |
|---|---|---|---|
| inch | 0.0254 | = | meter (m) |
| foot | 0.3048 | = | meter (m) |
| pound | 0.4536 | = | kilogram (kg) |
| foot-lb | 1.3558 | = | joule (j) |
| feet/sec | 0.3048 | = | meter per second (m/s) |
| Fahrenheit (°F) | (t°F - 32)/1.8 = t°C | | Celsius (°C) |

Note: Conversion factors can be associated with IEEE/ASTM SI-10 entitled "American National Standard for Use of the International System of Units (SI): The Modern Metric System"

6.15 Subject term (key word) listing.

Artillery
Barrels
Hammering
Hot work
Thermomechanical
Tubes

6.16 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - MC

Preparing activity:
Army - AR
(Project FORG-2026-003)

Review activities:
Army - AT, AV, CR4, MI, MR, TE
DLA – CQ

Civil Agency:
GSA - FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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Check the source to verify that this is the current version before use.